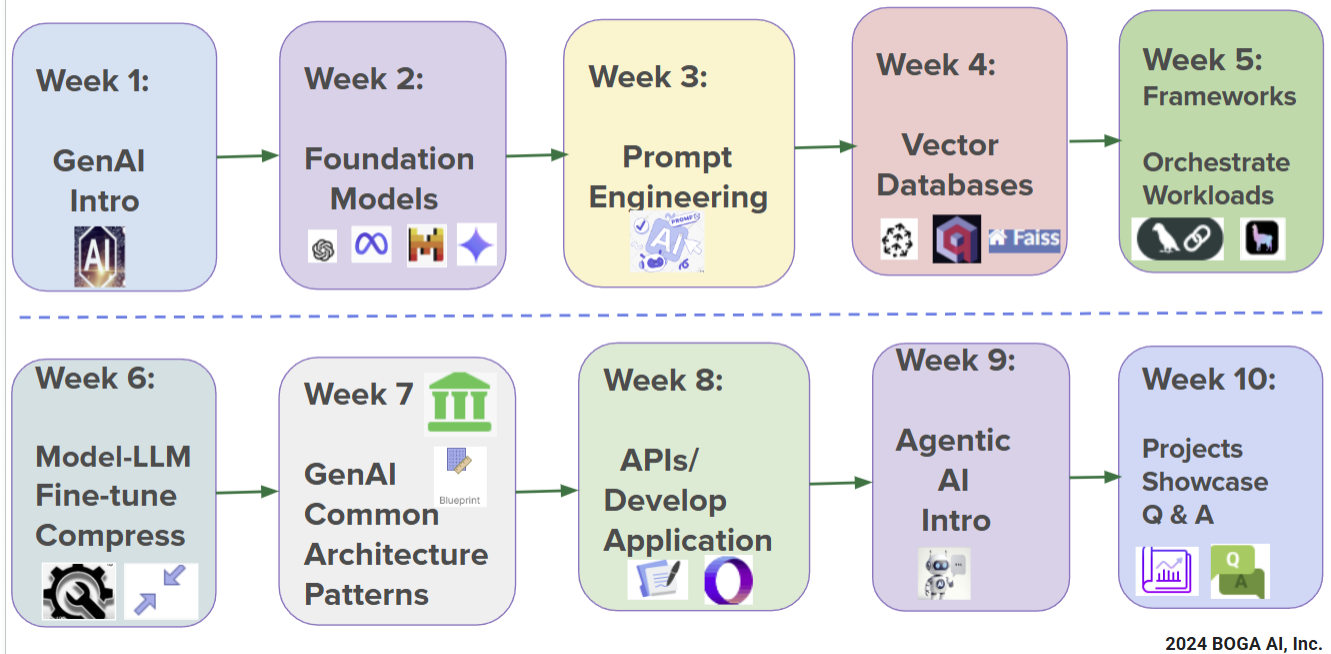


Generative AI

GenAI/Agentic AI Journey



Course Overview

Comprehensive course covering GenAI fundamentals to advanced topics that includes sharable notebooks and slides (# ~200).

- ❖ Live online sessions
- ❖ Hands-on projects
- ❖ Course fee: contact: bhoga01.ai@gmail.com or Cell: 323-274-5867
- ❖ Prerequisites: Python, Machine Learning fundamentals/MLOps.
- ❖ Schedule: Saturday/Sunday Evenings.

Week01: Introduction to Generative AI

- ❖ Discuss and describe what is AI and its branches.
- ❖ The evolution and scope of AI and its market insights.

- ❖ Briefly discuss about various traditional machine learning techniques and their purposes.
- ❖ Discuss about Predictive AI Practice: MLOps.
- ❖ Discuss about Generative AI Practice: LLMOps.
- ❖ Discuss difference between MLOps and LLMOps.
- ❖ Discuss Generative Use cases.
- ❖ Describe various new roles that will be created in Generative AI/Agentic AI world.
- ❖ Describe Generative AI Technology stack.
- ❖ Lab: Discuss how to setup and install a lab practice environment.
 - Signing up various cloud tools such as Google Colab, Google Cloud platform.
 - How to get API keys for Generative AI tools/LLMs.
 - How to setup local LLM runtime environment using Ollama.
 - How to setup a local ChatGPT like environment with web access.

Week02: Foundational Models:

- ❖ Define and describe what is an LLM and its capabilities.
- ❖ Discuss about different properties and characteristics of LLMs.
- ❖ Discuss how LLM generates tokens fundamentally.
- ❖ Discuss difference between regular LLMs and Reasoning LLMs with examples.
- ❖ Discuss about Open-source Model repository: **HuggingFace** and its purpose.
- ❖ Describe what is an embedding?
- ❖ Describe foundation models architecture: **Transformer** (self-attention, context and multi-headed attention)
- ❖ Discuss about various family of LLMs: OpenAI, Gemini, Llama, Mistral, OSS models like Falcon, Bloom and etc..
- ❖ **Lab:**
 - Introduction to “How to use setup and use google Colab environment?”
 - Demonstrate how to manage LLM API keys in Google Colab environment including how to setup google project.

- Demonstrate how to interact with various LLM providers like OpenAI, Groq, Gemini, Llama to get answers to user questions.
- Discuss about homework for week01 and week02.

Week03: Prompt Engineering:

- ❖ Define what is a prompt and what is the purpose of the prompt?
- ❖ Discuss the goal of Prompt engineering
- ❖ Discuss about Prompt Structure and its components with examples
- ❖ Discuss about Model parameters and its' importance.
- ❖ Design prompts for various Generative tasks
- ❖ Discuss different types prompts.
 - Zero-shot, One-shot, Few-shot, COT (chain of thought)
- ❖ Discuss few prompt tools and libraries to focus and learn more about prompts.
- ❖ **Lab:**
 - Demonstrate a various prompt technique with notebooks.

Week04: Vector databases:

- ❖ Define what is a vector database and the purpose of it?
- ❖ What problem the vector database solves it?
- ❖ How the data is represented in vector database/store?
- ❖ Differences between traditional database and vector database?
- ❖ Describe different components of vector database pipeline.
- ❖ Describe few popular vector databases (OSS) like Pinecone, FAISS, Qdrant, Chroma, Elastic Search
- ❖ **Lab:**
 - Demonstrate of ingesting a text/pdf file into vector store for retrieving context based on user query. (using Pinecone and FAISS)
 - Homework: Will get an opportunity to use a different vector store (like Qdrant) as knowledge for query retrieval.

Week05: Frameworks:

- ❖ Define what is a framework and its' role in Generative Application life cycle.
- ❖ Discuss about two popular unstructured/structured data processing frameworks i.e langchain and LlamaIndex
- ❖ Discuss about different types of LLM frameworks used in GenAI application development.
- ❖ Describe benefits of hosting LLMs at locally/on-premises.
- ❖ **Lab:**
 - How to download a LLM from huggingface?
 - How to deploy a LLM to GPU in notebook environment with Google Colab?
 - Describe how to define vector store as retrieval engine.

Week06: Compression/Quantization & Fine-tuning:

- ❖ Describe what kind of problem that Quantization and Fine-tuning solves?
- ❖ Discuss few Quantization and Fine-tuning techniques (GPTQ, AWQ, LoRA/QLoRA)
- ❖ **Lab:**
 - Demonstrate how to apply Quantization and other techniques like LoRA and QLoRA)

Week07: GenAI Usage Architectures:

- ❖ Describe a common Generative AI Architecture pattern.
- ❖ Describe what is an Information Retrieval - IR and its use case.
- ❖ Describe what is a Retrieval-Augmented Generation - RAG (IR + G) and its use case.
- ❖ Describe what is an Adv... RAG with reRanker (CrossEncoder Model:) and its use case.
- ❖ Describe in detail what is a Hybrid Search (BM25 + Semantic:KNN)
- ❖ Discuss in detail what a Contextual RAG is about and its use case?
- ❖ What are GraphRAG (MS) & LightRAG about? How different this Architecture from RAG and what use cases it solves?
- ❖ Describe what is a Production RAG Pipeline looks like?
- ❖ **Lab:**

- Demonstrate Hybrid Search and Contextual RAG using Notebooks with Pinecone Database?
- Demonstrate LightRAG example using notebook.

Week08: APIs: Development of REST APIs

- ❖ Discuss about Generative AI software ecosystem landscape.
- ❖ Discuss about top 20 python libraries for every AI engineer to know.
- ❖ Describe a generic use case implementation Architecture and its components.
- ❖ Define what is an API about and how to define a python function as an API endpoint?
- ❖ **Lab:**
 - Demonstrate how to create REST API using FastAPI and invoke it from another python program or using Postman tool.
 - Demonstrate a simple RAG application using Streamlit App.
 - Demonstrate a simple Text to SQL application using Streamlit App.
 - Demonstrate how to track end-2-end Generative AI application for debugging and tracing purposes using Arize observability tool.

Week09: Agentic AI: Introduction:

- ❖ Describe what is an agent and how is it different from chatbot
- ❖ Discuss about different use cases that Agentic AI solves.
- ❖ Describe components of Agent
- ❖ Discuss about a common Architecture patterns
- ❖ Discuss about few common Agentic frameworks.
- ❖ **Lab:**
 - Demonstrate a simple agent using LangGraph framework.

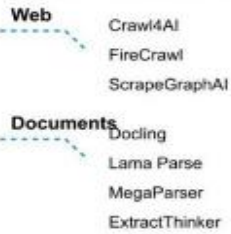
Week10: Projects & Q&A:

- ❖ You will get a chance to present a use case/project that is done as part of training.
- ❖ Q & A /Next steps

❖ Targeted Tech-stack.

Extraction

Pull structured and unstructured data from different sources



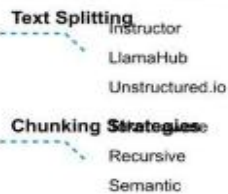
Text Embeddings

Convert text into embeddings for similarity search



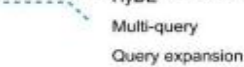
Data Processing & Chunking

Process and segment documents for optimal retrieval



Query Understanding

Transform queries for better retrieval precision



Retrieval Enhancement

Improve relevance of retrieved documents



Vector Databases

Store and search vector embeddings



Unified Semantic Space

Unified Semantic Space

LLMs

Handle generation using retrieved context



Knowledge Graphs

Structure and connect information



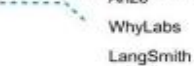
Open LLMs Access

Enable running open models locally



Observability

Monitor and debug your RAG pipeline



Frameworks

Simplify development so you don't have to build everything from scratch



Orchestration

Orchestration



Evaluation

Evaluate RAG System performance

